Renal and Genitourinary Considerations in Pediatrics Part II

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Thomas Chan D.O.
Associate Professor
Dept. of Clinical Specialties
tchan02@nyit.edu



Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

- Understand the important clinical aspects of a child suspected of having a GU problem
- Discuss the work up, treatment and management of a child suspected of having a GU problem
- Discuss various systemic disease states in children and implications to the renal system
- Discuss various structural problems in the GU system in a child and their clinical significance
- Understand enuresis in children and the various treatment modalities

Source: Course Syllabus

When to suspect kidney issues?

- Abnormal urine by history
- ***Abnormal urinalysis***
- Abnormal exam- peri-orbital swelling, generalized swelling, scrotal edema
- Hypertension
- Fever
- Flank pain

Hematuria

- Gross Hematuria?
- Trauma?
- Back pain?
- Fever?
- Menses?
- Family History?

- Work up?
  - Rpt UA
  - Urine Cx
  - Complete metabolic Panel (BUN/Cr)
  - Spot Ca/Creatine Ratio (<0.2nl)
  - 24hr urine for Ca/Cr
  - RUS
  - Other imaging:CT scan/ MRI
  - Nephrology consult
  - Urology consult

Hematuria in Children

Microscopic/Gross

- Relatively Benign
  - Exercise induced
  - Trauma related
  - Benign familial
  - Sickle trait/Dz
  - Stones/ hypercalciuria
  - UTI

OTHER CAUSES

- Alport syndrome
- PSGN
- IgA nephropathy- Berger's (intermittent micro and gross)
- Glomerulonephritis
- Lupus
- Tumors
- Other Systemic illness

Proteinuria in children

Abnormal Urinalysis-asymptomatic

- Rpt with first morning Voided UA
- Spot Protein/Creatine ratio (<0.2nl)
- 24 hr urine Protein/Cr





- 5year old girl presents to the office with swollen eyes noticed upon waking up. Mom is worried about allergies. Goes away later in the day



Nephrotic syndrome

Findings

- ***Edema*** (AM peri-orbital edema)
- Proteinuria
- Hypoalbuminemia
- Hyperlipidemia





Treatment/ Workup

Likely **Minimal Change Dz**

- Corticosteroids

- When to Biopsy?
  - Recurrent relapsers
  - Steroid resistant
  - Nephritis (HTN)
- Dietary- NI protein, low salt

Nephrotic Syndrome- Watch out for…

Infections

- Fever and abdominal pain
  - **Peritonitis** (Strep Pneumoniae)
- Sepsis, pneumonia/ cellulitis

Hypercoagulable state

12 year old male

Parents says his face looks swollen and he is not feeling well. Very tired per parents

He reports his urine looks very dark

Parents say he was sick 2 weeks ago with fever and a sore throat and recovered on own. Never saw a physician for illness.

Vitals: HR 98, rr 20, BP 138/ 98, rpt 136/100,
Pox 99% RA

Post Streptococcal Glomerulonephritis (PSGN)

- Sudden onset of gross hematuria
- Edema
- Hypertension
- Oliguria
- Low Complement level
- Evidence of strep
  - ThCx/ Rapid strep
  - Antistreptolysin O titers (ASO)
  - Anti-DNAse B Antibody



Post Streptococcal Glomerulonephritis (PSGN)

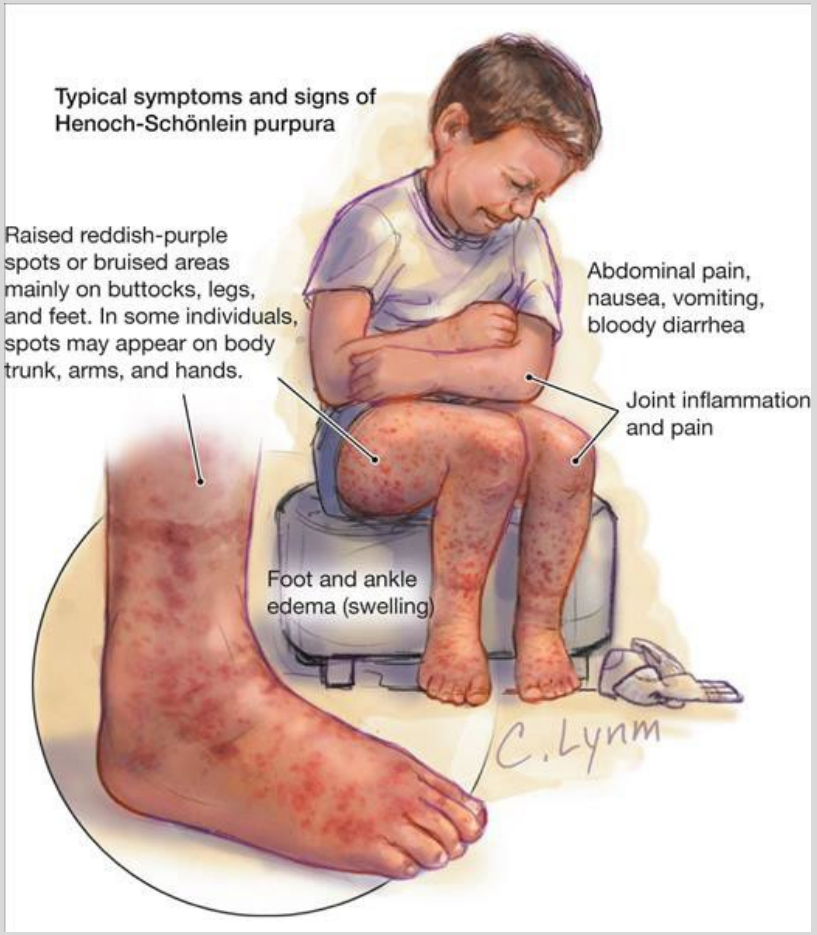
- Renal insufficiency
- Decreased GFR
- Fluid Restriction
- Acute phase done by 6-8 weeks

- Complement level (C3) should return to nl in 6-8 weeks
- Micro Hematuria may persist 1-2 yrs
- Antibiotics do not eliminate risk (does for rheumatic fever)

- 4 year old male presents with abdominal pain.
  - Recent viral illness 2 weeks ago and resolved. Parent says he is not using his right hand as much.
  - On exam you note a palpable non blanching blotchy rash to the elbows and buttocks and back of legs

Henoch-Schonlein Purpura

- Fever
- Joint Pain
- Abdominal pain
- Palpable Purpura
- No thrombocytopenia!
- Sterile pyuria



Henoch-Schonlein Purpura



Scrotal edema



- IgA deposits
- Renal Failure is possible
- Association with Intussusception
- Steroids may help symptoms

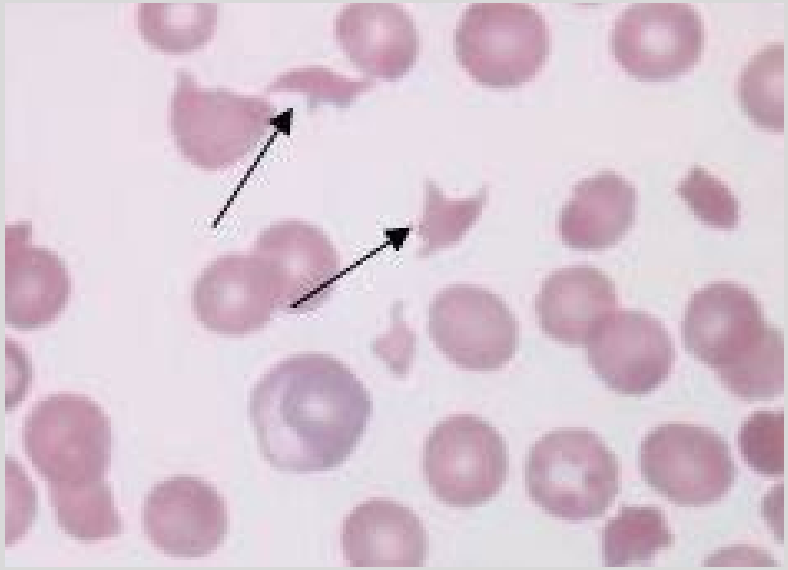
- 10 year old was at BBQ over the weekend. Now fever, abdominal pain, rash, vomiting, bloody stool



Hemolytic Uremic Syndrome

- Hemolysis
  - Microangiopathic Anemia- Schistocytes
  - High Reticulocyte count
- Uremia
  - Oliguria
  - High BUN/Cr, proteinuria/hematuria/casts
- Petechiae/Purpura
  - Thrombocytopenia
- Bloody Diarrhea
  - Fever

- Neurological (less common)
  - Confusion
  - Irritable
  - Stroke
  - Relation with TTP



Hemolytic Uremic Syndrome

Infection

*** **Ecoli 0157-H7** – shigatoxin***

- Shigella
- StrepPneumonia
- Drug Toxicity

Treatment- supportive care if Ecoli (dialysis, IVF, transfusions)

- Antibiotics?...NO!
  - Potential to cause more damage by increasing toxin load

OUTBREAKS COMMON: Contaminated food and drink:

improperly cooked foods, contaminated drinking water, produce,
unpasteurized milk/ cheese

- 8 year old male presents to the office with abdominal pain. You notice him looking uncomfortable walking hunched over holding lower abdomen. Physical exam including Abdominal exam is benign….

You check his genitalia to complete the exam and….find it is his genital area that is hurting him



Diff Diagnosis of **Scrotal Pain**

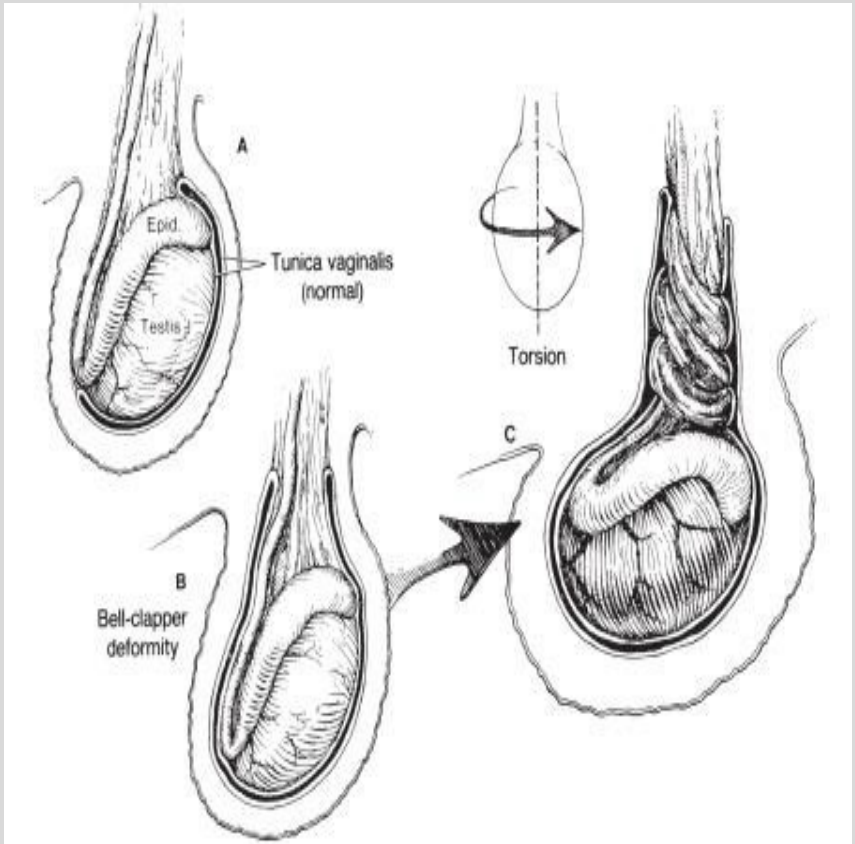
- Testicular Torsion
- Incarcerated Hernia
- Acute Epididymitis
- Orchitis
- Torsion of appendix testes
- Trauma
- Other…



Dx Testicular torsion- race against the clock!

Always check testicles in abdominal or groin pain!!!!!

- Swollen testicle
- High riding
- Only have 6-8hrs from onset of pain

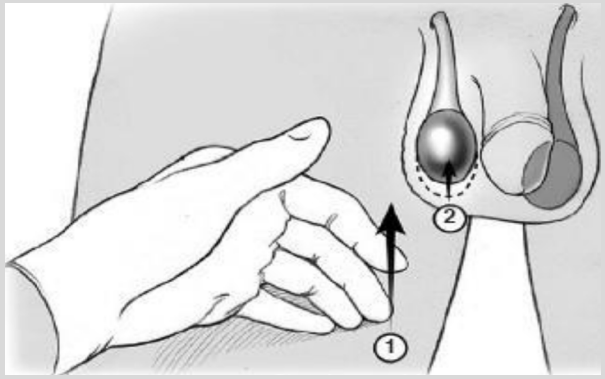


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Testicular Torsion- Crematerіc and Prehn



Figure 3. Cremasteric Reflex



The re ex is elicited by (1) stroking the ipsila eral inner high with a tongue depressor or gloved hand, resulting in (2) the elevation of he testicle throughcontrac on of the cremasteric muscle.
From: Davis JE. Chapter 31: Scrotal Pain, in Mahadevan Sli, *Garmel GM, eds. An Introduction to Clinical Emergency Medicine: Guide for Prac itioners in the Emergency Department. New York: Cambridge University Press; 2005:p466 . Reprinted with the permission of ca m bridge Universi ty Press.*

- **Absent cremasteric reflex**

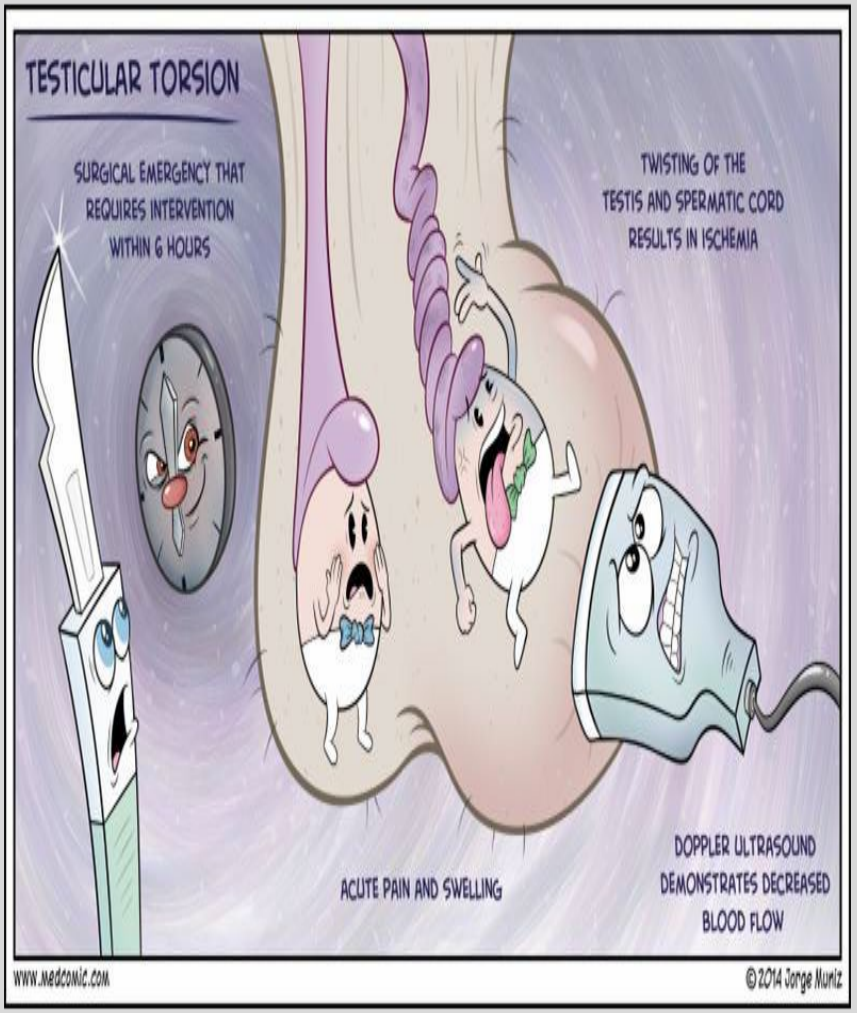
Prehn Sign

- -ve Prehn's sign indicates no pain relief with lifting the affected testicle, which points towards testicular torsion which is a surgical emergency and must be relieved within 6 hours.
- Positive Prehn's sign indicates there is pain relief with lifting the affected testicle, which points towards epididymitis.

- **Negative/Absent Prehn sign (unreliable)**

Testicular torsion

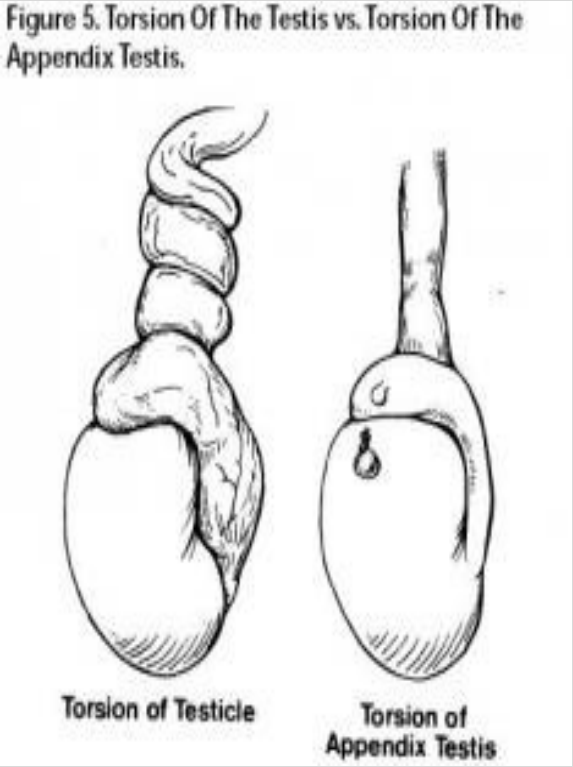
- ***Testicular US with doppler***
- Radionuclear scan
- Urology- orchiopexy BOTH sides



Torsion of Appendix testis

- **Blue dot sign**
- Give NSAID's for pain
- Reassure





Torsion of appendix testes

Ovarian Torsion

- Severe acute, abdominal pain
- Nausea
- Vomiting

Index of suspicion to make the Dx
Surgical correction



Imperforate Hymen

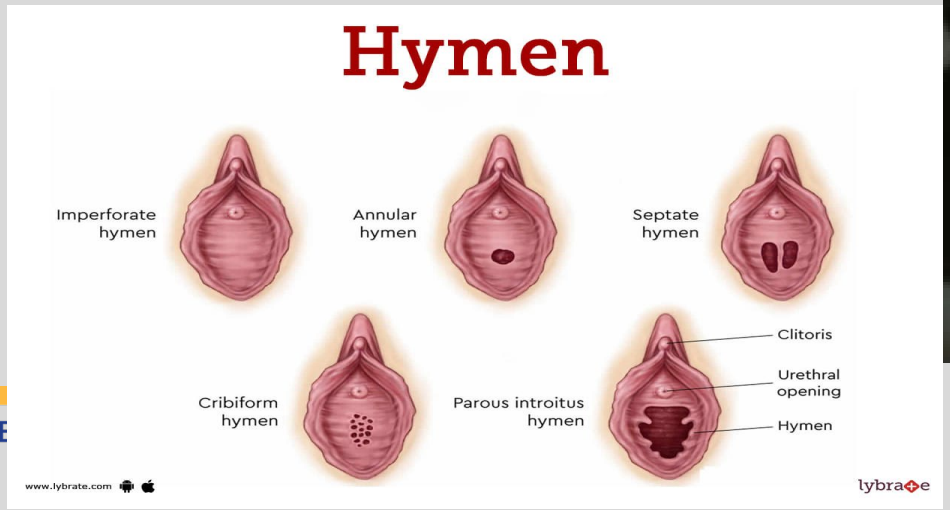
Often unrecognized till adolescence, sometimes recognized as a whitish bulge in neonates

Adolescent with recurrent monthly abdominal pain, back pain, urinary issues- can be severe over time and cause infertility issues, hydronephrosis/ kidney damage (mechanical obstruction of urinary tract)

Bulging hymen- often bluish from blood (hematocolpos)

Need surgical correction- Hymenectomy





Nocturnal Enuresis



- 8 year old boy is wetting the bed almost every night. He has always wet the bed and parents thought he would grow out of it but never has. Otherwise well. Parents are frustrated and child is embarrassed.

Primary Enuresis

- Never got full control of bladder
- 'Problem' after 6 years old
- Family Hx -similar in parent

8 year old boy/ girl

Is wetting the bed almost every night. This is new for the last 2 weeks. Was completely 'dry' and toilet trained by 4-5 yrs old.

Secondary Enuresis

- Was fully toilet trained and now lost control
- Psycho-social stressors predominant reason
- R/o other causes (UTI)

Treatment in **Primary** Nocturnal Enuresis

- **Fluid Restriction**
- **Bed Wetting alarm**
- **Reward Calendar**
- No Fault/ No Blame
- Biofeedback Psychotherapy
  - Reward
  - Brain over bladder
  - Help clean
- Medications
  - **Vasopressin(DDAVP)**
  - Tricyclic Antidepressants- rarely used now



Bed Wetting Alarm and Reward Calendar







Summary Slide 1

- Early morning puffy, swollen eyes
- HTN in children think kidneys
- Hematuria- send Ca/Creatine ratio
- Proteinuria send Protein/ Creatine ratio
- Post Strep GN- C3 low should return to nl by 6-8 weeks, if not look for other reason
- Non blanching *rash to buttocks*, joints, fever, abdominal pain- HSP
- Hemolytic Uremic Syndrome- hemolysis (anemia), uremia (dark or decreased urine, abnormal BUN/Cr), syndrome (fever, bloody stools, systemic complaints) –Ecoli 0157- Don't give antibiotics

Summary Slide 2

- Always check testicles in Abdominal pain/ Inguinal pain
- Testicular torsion- get Ultrasound with doppler

  (You only have 6-8 hrs from start to save the testicle)
- Blue dot sign- torsion of Appendix testes- Reassurance
- Abdominal/ pelvic pain in adolescent female- consider ovarian torsion in DiffDiag, if recurrent/monthly ab pain and no menses- think imperforate hymen
- Secondary enuresis- think psychosocial issue
- Primary enuresis- Bed wetting alarm, reward

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R